```
;clc
;clear
;close all

=========================================================%
Faulhaber Polynomial Coefficients %
Using Vandermonde Matrix %
=========================================================%

;Kmax = 20

---------------------------------------------------------%
Matrix to store coefficients %
Row k contains coefficients of power k %
---------------------------------------------------------%
;CoeffMatrix = sym(zeros(Kmax,Kmax+1))

for k = 1:Kmax

;fprintf('Computing k = %d\n',k)

;N = k + 1

-----------------------------------------------------%
Construct Vandermonde Matrix %
-----------------------------------------------------%
;A = sym(zeros(N,N))
```

```
for m = 1:N
for j = 1:N
;A(m,j) = sym(m)^(N-j+1)
end
end

----------------------------------------------------%
Construct RHS %
----------------------------------------------------%
;b = sym(zeros(N,1))

for m = 1:N

;s = sym(0)

for r = 1:m
;s = s + sym(r)^k
end

;b(m) = s

end

----------------------------------------------------%
Solve the system %
----------------------------------------------------%
;coeff = A\b
```

```
----------------------------------------------------%
Store coefficients in the matrix %
----------------------------------------------------%
;'.CoeffMatrix(k,1:N) = coeff

----------------------------------------------------%
Display coefficients %
----------------------------------------------------%
;fprintf('\n')
;fprintf('=======================================\n')
;fprintf('Power k = %d\n',k)
;fprintf('=======================================\n')

for i = 1:N
;fprintf('%s\n',char(coeff(i)))
end

end

=========================================================%
Display complete matrix %
=========================================================%

(' ')disp
('=======================================')disp
disp('Matrix of Faulhaber Coefficients')
('=======================================')disp
```

```
disp(CoeffMatrix)
==========================================================%%
=
Version 1 : Column Analyzer %%
==========================================================%%
=

(' ')disp
('=========================================')disp
disp('COLUMN ANALYSIS')
('=========================================')disp

syms k

;[numRows,numCols] = size(CoeffMatrix)

for col = 1:numCols

----------------------------------------%
Ignore zero columns %
----------------------------------------%
;idx = find(CoeffMatrix(:,col)~=0)

if length(idx)<2
continue
end

;x = idx
;y = CoeffMatrix(idx,col)
```

```
----------------------------------------%
Polynomial interpolation (Lagrange) %
----------------------------------------%
;P = sym(0)

;n = length(x)

for i=2:n

;L = sym(1)

for j=2:n

if i~=j
;L = L*(k-x(j))/(x(i)-x(j))
end

end

;P = expand(P+y(i)*L)

end

;P = simplify(P)

----------------------------------------%
Degree %
```

```
----------------------------------------%
;d = feval(symengine,'degree',P,k)

----------------------------------------%
Factorization %
----------------------------------------%
;F = factor(P)

----------------------------------------%
Roots %
----------------------------------------%
;R = solve(P==0,k)

----------------------------------------%
Leading coefficient %
----------------------------------------%
;a = coeffs(P,k,'All')
;LeadCoeff = a(1)

----------------------------------------%
Display %
----------------------------------------%
;fprintf('\n')
;fprintf('====================================\n')
;fprintf('Column %d\n',col)
;fprintf('====================================\n')

disp('Polynomial:')
```

```
disp(P)

disp('Degree:')
disp(d)

disp('Factorization:')
disp(F)

disp('Roots:')
disp(R)

disp('Leading coefficient:')
disp(LeadCoeff)

end
```